

20GNS502J Research Methodology

Unit / Module IV: Data Collection, Analysis and Inference

Part B

M. Tech. Coursework

SRM Institute of Science and Technology

Unit / Module IV: Data Collection, Analysis and Inference

- Basic probability distributions – types – applications in Engineering – statistics – types
- Normal distribution – exercises –problem solving using software tools
- Binominal and Poisson distributions – exercises using software
- Weibull and exponential distributions
- Sampling: types – size of sample – sample designs
- Sampling tests: Student t-test – applications in engineering / research – exercises using software tools
- F-test and its application in research studies – exercises using software
- χ^2 test and its application in research studies – exercises using software
- Correlation and regression analysis – time series analysis
- Forecasting methods
- Factor analysis, cluster analysis and discriminant analysis
- Design of experiments - basic experimental designs: completely randomized design
- Randomized block design, Latin square design - exercises
- Full factorial design - 2^2 , 2^3 and 2^4 only – exercises
- Fractional factorial designs – exercises
- Accuracy, precision and error analysis

Unit / Module IV: Data Collection, Analysis and Inference

Topics (12 Hours)

- Basic probability distributions – types – applications in Engineering – statistics – types
 - Normal distribution – exercises – problem solving using software tools
 - Binominal and Poisson distributions – exercises using software
 - Weibull and Exponential distributions
- Sampling: types – size of sample – sample designs
- Sampling tests:
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 - F-test and its application in research studies – exercises using software
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Unit / Module IV: Data Collection, Analysis and Inference

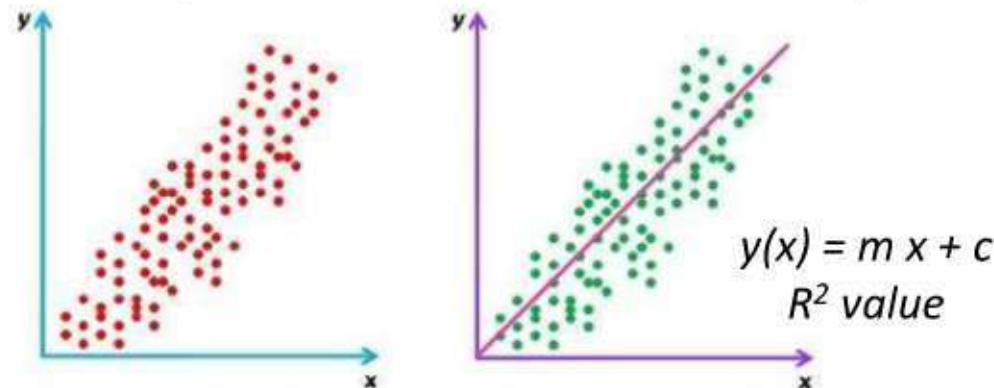
Topics (12 Hours)

- Correlation and Regression analysis – time series analysis
- Forecasting methods
- Factor analysis, cluster analysis and discriminant analysis
- Design of experiments - basic experimental designs:
 - Completely randomized design
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 - Fractional factorial designs – exercises
- Accuracy, precision and error analysis
- Project

Correlation and Regression

Establishing nature of relationship between related variables

- **Correlation** (Co' (together) and relation (connection)) is described as the analysis which describes the association or the absence of any relationship between variables. In the picture below for $y = f(x)$, as variable x increases, the variable y also increases, hence a positive relationship appears to exist.
- **Regression** analysis, predicts the value of the dependent variable based on the known value of the independent variable, assuming that average mathematical relationship between two or more variables (with the trend line and equation as shown).



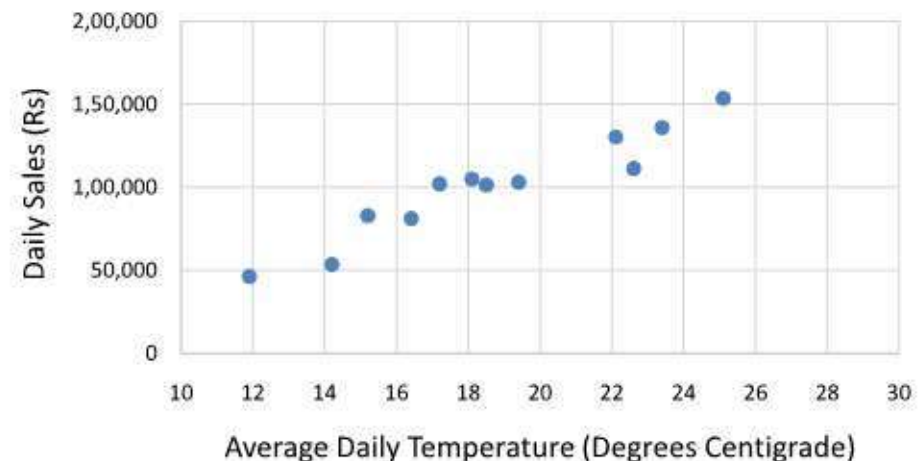
Correlation Vs Regression

Correlation - Example

- The local ice cream shop keeps track of daily ice cream sales against the average temperature on the day.
- The data for first 12 days in March 2023 is tabulated.
- A hypothesis exists that sales increases with increase in the temperature.
- The scatter plot shows that the hypothesis is valid with warmer weather leading to higher sales.
- There is a positive correlation here (both sales and temp increase together), but not a perfect one!

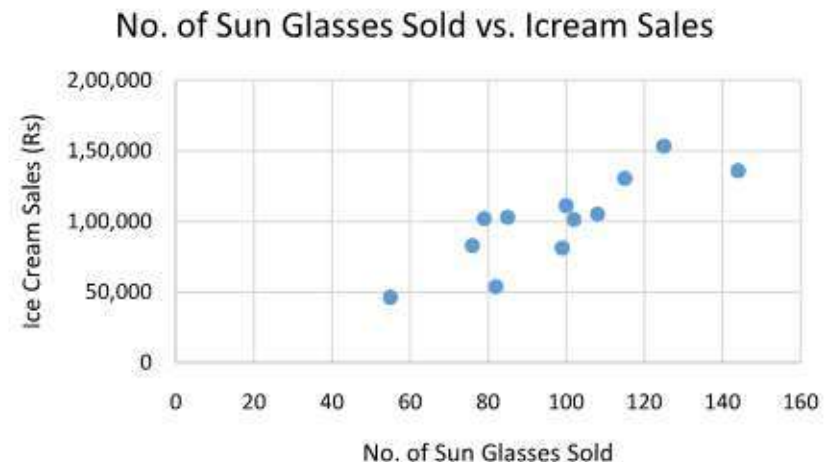
Date in March 2023	Avg. Daily Temp. (°C)	Daily Ice Cream Sales (Rs)
1	14.2	53,750
2	16.4	81,250
3	11.9	46,250
4	15.2	83,000
5	18.5	101,500
6	22.1	130,500
7	19.4	103,000
8	25.1	153,500
9	23.4	136,000
10	18.1	105,250
11	22.6	111,250
12	17.2	102,000

Temperature vs. Ice Cream Sales in March 2023



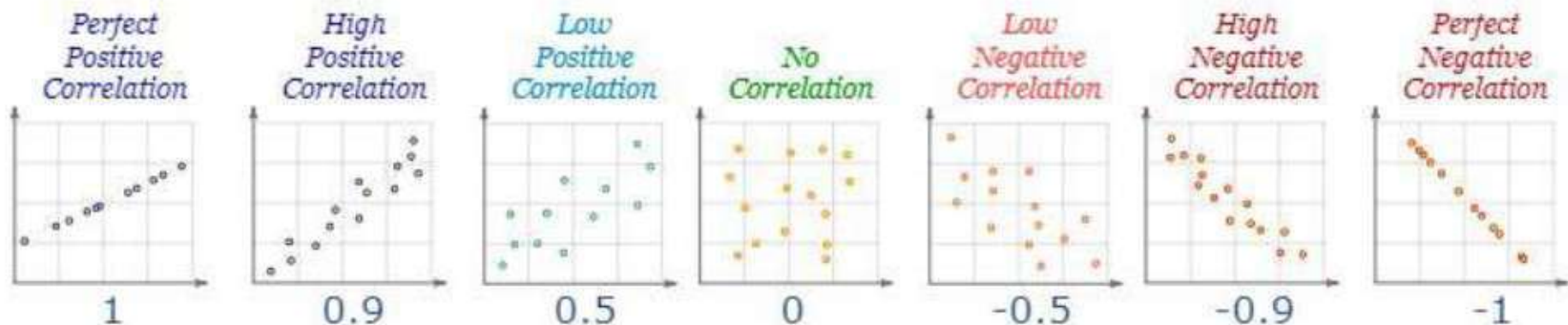
Correlation is not Causation

- What appears to be a correlation between y and x variables, does not in any way prove that y is dependent on x . So, x need not cause y . For *e.g.*:
 - y is the independent variable and the causal factor for x – other way around
 - y and x may be linked to different causal factor
 - What appears as correlations might be due to a random chance!
- Let us take the example of the ice cream sales again and attempt to correlate it with the sales of next-door sunglasses shop as shown
- The plot shows a positive correlation suggesting that ice cream sales is related to sunglass sales, which does not make sense!
- However, both are related to the temperature as sunglass as well as ice cream sales are both dependent on the temperature
- So, sunglass sales does not directly influence ice cream sales



Correlation Coefficient

- A Correlation is Positive when the values increase together, and is Negative when one value decreases as the other increases
- A Correlation is assumed to be linear (following a trend line) and the **correlation coefficient** can have values ranging from -1 to +1.
 - +1 correlation coefficient is a perfect positive correlation
 - 0 is no correlation (the values don't seem linked at all)
 - -1 correlation coefficient is a perfect negative correlation
- The value shows how good the correlation is (not how steep the line is), and if it is positive or negative.



Types of Correlation and Simple Correlation

- There are three types of Correlations:
 1. Simple Correlation
 2. Multiple Correlation
 3. Partial Correlation
- **Simple Correlation:** The study of a relationship between two variable only, *i.e.* a dependent and an independent variable is known as simple correlation
- For e.g., $y = f(x)$, where y is the dependent and x is the independent variable as shown with examples below

y (Dependent Variable)	x (Independent Variable)
Savings in a Family	Income
Profit of a Company	Sales
Agricultural Produce Yield	Rainfall

- The savings of family is dependent on the income showing simple one-is-to-one relationship

Simple Correlation Coefficient

- The Correlation coefficient for simple correlations can be calculated with the formulae:

$$R_{xy} = \frac{\frac{\sum xy}{n} - \bar{x} \bar{y}}{\sqrt{\frac{\sum x^2}{n} - \bar{x}^2} * \sqrt{\frac{\sum y^2}{n} - \bar{y}^2}}$$

- Let us take the example of the ice cream sales to tabulate and calculate this:

- $n = 12$
- $\bar{x} = 18.675$
- $\bar{y} = 1,00,604.2$
- $\frac{\sum x^2}{n} = 363.5$
- $\frac{\sum y^2}{n} = 11,03,13,80,208.3$
- $\frac{\sum xy}{n} = 1989720.8$
- Substituting these values:
- $R_{xy} = 0.9576$ (matches with the spreadsheet calculation)

Date in March 2023	Avg. Daily Temp. (°C) (x)	Daily Ice Cream Sales (Rs) (y)	x^2	y^2	xy
1	14.2	53,750	202	2,889,062,500	763250
2	16.4	81,250	269	6,601,562,500	1332500
3	11.9	46,250	142	2,139,062,500	550375
4	15.2	83,000	231	6,889,000,000	1261600
5	18.5	101,500	342	10,302,250,000	1877750
6	22.1	130,500	488	17,030,250,000	2884050
7	19.4	103,000	376	10,609,000,000	1998200
8	25.1	153,500	630	23,562,250,000	3852850
9	23.4	136,000	548	18,496,000,000	3182400
10	18.1	105,250	328	11,077,562,500	1905025
11	22.6	111,250	511	12,376,562,500	2514250
12	17.2	102,000	296	10,404,000,000	1754400
Total	224.1	1,207,250	4,362	132,376,562,500	23,876,650
Mean	18.7	100,604.2	363.5	11,031,380,208.3	1,989,720.8
Corr. Coeff.	0.9575				

Simple Correlation Coefficient

- Raw data for the number of preparation hours against the marks scored in a test by various students of a class are tabulated below. Calculate the correlation coefficient manually for the number of hours of preparation against the marks scored and then confirm with MS Excel. (Answer: $R_{xy} = 0.704$)

	Student's	
	Preparation Hours	Marks Scored
1	8	98
2	2	74
3	6	87
4	4	82
5	2	7

Perfect Positive and Negative Errors in Correlation

Positive and Negative Errors

- The change in magnitude of the dependent and independent variable are in the same direction for **positive errors** and *vice-versa* as shown in the adjacent table

Perfect Positive and Negative Errors

- The change in magnitude of the dependent and independent variables are constant and equal in the same direction for **perfect positive errors**, but opposite in direction as shown in the adjacent table for **perfect negative errors**

Positive Error		Negative Error	
y	x	y	x
1	2	2	110
5	6	8	88
8	15	10	16
10	22	17	4
12	45	22	2
As y increases x also increases		As y increases x decreases	
Perfect Positive Error		Perfect Negative Error	
y	x	y	x
1	2	2	9
5	6	4	7
8	15	6	5
10	22	8	3
12	45	10	1
$\Delta(x-y) = \text{Constant} = 1$		$\Delta y = -\Delta x$	
Also		and Constant = 1	
$\Delta x = \Delta y = \text{Constant} = 4$			

Multiple Correlation Coefficient

- The coefficient of multiple correlation measures the relationship between a dependent variable and a host of independent variables coefficient taken together
- Let x_1, x_2, x_3 be variables with x_1 being dependent variable and x_2 and x_3 being independent variables wherein $x_1 = f(x_2, x_3)$
- Possible examples of this scenario are:

Dependent Variable	Independent Variables	
y	x_1	x_2
Savings in a Family	Income	Size of family
Profit of a Company	Sales	Advertising Expenses
Agricultural Produce Yield	Rainfall	Soil Fertility

Multiple Correlation Coefficient

- Let us take the example of Profit = f(Sales, Advertisement Expenses) for a company as an example and let suffixes 1, 2 and 3 denote Profit, Sales and Advertisement Expenses respectively.
 - r_{12} = relationship between Profit and Sales
 - r_{13} = relationship between Profit and Advertisement Expenses
 - r_{23} = relationship between Sales and Advertisement Expenses
- The multiple correlation coefficient, R is given by:

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{23}}{1 - r_{23}^2}} \text{ with } 0 \leq R_{1.23}^2 \leq 1$$

- Note: Symmetry relationship exists with

$$R_{12} = R_{21} ; R_{13} = R_{31} \text{ and } R_{23} = R_{32}$$

Partial Correlation Coefficient

- The coefficient of partial correlation measures the relationship between a dependent variable and an independent variable, while keeping the rest of the independent variables constant
- Let x_1, x_2, x_3 be variables with x_1 being dependent and x_2 being the independent variable under consideration with x_3 being held constant such that it can be depicted as $x_1 = f(x_2 | x_3)$
- Possible examples of this scenario are:

Dependent Variable	Independent Variables	
y	x_1	x_2
Savings in a Family	Income	Size of family
Profit of a Company	Sales	Advertising Expenses
Agricultural Produce Yield	Rainfall	Soil Fertility

- Here, the question is what is the influence of income on the savings of a family while keeping the family size constant...

Partial Correlation Coefficient

- Let us take the example of $Profit = f(Sales, Advertisement Expenses)$ for a company as an example and let suffixes 1, 2 and 3 denote Profit, Sales and Advertisement Expenses respectively.
 - r_{12} = relationship between Profit and Sales
 - r_{13} = relationship between Profit and Advertisement Expenses
 - r_{23} = relationship between Sales and Advertisement Expenses
- For the case of influence of sales on the profit of a company while maintaining the advertisement constant $\{x_1 = f(x_2|x_3)\}$ the partial correlation coefficient, R is given by:

$$R_{12.3} = \frac{R_{12} - R_{13} R_{23}}{\sqrt{1 - R_{13}^2} \sqrt{1 - R_{23}^2}} \text{ with } 0 \leq R_{12.3}^2 \leq 1$$

Multiple Correlation Coefficient - Assignment

1. From the following data, obtain the multiple correlation coefficients, $R_{1.23}$, $R_{2.13}$ and $R_{3.12}$.

X1	2	5	7	11
X2	3	6	10	12
X3	1	3	6	10

(Answers: $R_{1.23} = 0.99$, $R_{2.13} = 0.97$ and $R_{3.12} = 0.99$)

Regression Analysis

- Regression analysis, predicts the value of the dependent variable based on the known value of the independent variable, assuming that a mathematical relationship exists between them (e.g., $y = mx + c$) in the form of $Income = m(Sales) + c$.
- Regression is also used to estimate the strength of the relationship between two variables (e.g., the income and sales of an company).

Simple Regression Analysis

- It is used to estimate the relationship between variable, one dependent and the other the independent variable
- Let us take the example of $y = f(x)$. To predict the value of y for a given x value, the relationship needs to be established which requires the collection of a sample data of size, say “ n ”.
- Using this data the regression coefficient can be determined by tabulating Σy_i , Σx_i , $\Sigma x_i y_i$, Σx_i^2 and Σy_i^2 as shown in the next slide.

Simple Regression Analysis

y = dependent and x = independent variable					
n_i	y	x	xy	x^2	y^2
1	y_1	x_1	$x_1 y_1$	x_1^2	y_1^2
2	y_2	x_2	$x_2 y_2$	x_2^2	y_2^2
	-	-	-	-	-
	-	-	-	-	-
n	y_n	x_n	$x_n y_n$	x_n^2	y_n^2
Total	Σy_i	Σx_i	$\Sigma x_i y_i$	Σx_i^2	Σy_i^2
Mean	$\Sigma y_i / n$	$\Sigma x_i / n$	$\Sigma x_i y_i / n$	$\Sigma x_i^2 / n$	$\Sigma y_i^2 / n$

Regression line of y on x can be obtained from the expression:

$$(y - \bar{y}) = b_{y_x} (x - \bar{x})$$

$\bar{x}$ and $\bar{y}$ are mean of x and y respectively given by:

$$\bar{x} = \frac{\Sigma x_i}{n} \text{ and } \bar{y} = \frac{\Sigma y_i}{n}$$

Where Regression Coefficient of y on x = $b_{y_x} = \frac{\frac{\Sigma xy}{n} - \bar{x}\bar{y}}{\frac{\Sigma y^2}{n} - \bar{y}^2}$

So if x is known, then unknown y can be calculated.

Simple Regression Analysis

y = dependent and x = independent variable					
n_i	y	x	xy	x^2	y^2
1	y_1	x_1	$x_1 y_1$	x_1^2	y_1^2
2	y_2	x_2	$x_2 y_2$	x_2^2	y_2^2
	-	-	-	-	-
	-	-	-	-	-
n	y_n	x_n	$x_n y_n$	x_n^2	y_n^2
Total	$\sum y_i$	$\sum x_i$	$\sum x_i y_i$	$\sum x_i^2$	$\sum y_i^2$
Mean	$\sum y_i / n$	$\sum x_i / n$	$\sum x_i y_i / n$	$\sum x_i^2 / n$	$\sum y_i^2 / n$

Similarly Regression line of x on y can be obtained from the expression:

$$(x - \bar{x}) = b_{xy} (y - \bar{y})$$

Where Regression Coefficient of x on y = $b_{xy} = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\frac{\sum x^2}{n} - \bar{x}^2}$

So if y is known, then unknown x can be calculated.

Note: If $b_{yx} > 1$, then $b_{xy} < 1$ and if $b_{yx} < 1$, then $b_{xy} > 1$

Simple Regression Analysis – Example 1

Problem: Calculate the regression coefficient and obtain the lines of regression for y on x the following data.

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

Solution: This problem involves only two variable, x and y . This would require a simple linear regression equation to be fitted.

$$\text{Slope of } y \text{ on } x = b_{yx} = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\frac{\sum x^2}{n} - \bar{x}^2} =$$

$$\frac{47.7 - (4.0 * 11.0)}{20.0 - (4.0 * 4.0)} = 0.925$$

The intercept is calculated from

$$\bar{y} = m\bar{x} + C \Rightarrow 11 = 0.925(4) + C \\ \Rightarrow C = 7.3$$

The Regression equation is:

$$y = 0.925x + 7.3$$

n	x	y	x^2	y^2	xy
1	1	9	1	81	9
2	2	8	4	64	16
3	3	10	9	100	30
4	4	12	16	144	48
5	5	11	25	121	55
6	6	13	36	169	78
7	7	14	49	196	98
Total	28	77	140	875	334
Mean	4.0	11.0	20.0	125.0	47.7

Simple Regression Analysis – Example 1

Correlation coefficient (R) is calculated using the formula

$$R_{xy} = \frac{\frac{\sum xy}{n} - \bar{x} \bar{y}}{\sqrt{\frac{\sum x^2}{n} - \bar{x}^2} * \sqrt{\frac{\sum y^2}{n} - \bar{y}^2}}$$

$$R_{xy} = \frac{47.7 - (4.0 * 11.0)}{\sqrt{20.0 - 4^2} * \sqrt{125 - 11^2}} = 0.925$$

The coefficient of determination $R_{xy}^2 = 0.856$

This implies that 85.6% of the variation in y with changes in x can be explained by the equation $y = 0.925x + 7.3$

We can also state that linear equations are reasonably the best fit as $R_{xy} = 0.925$

Simple Regression Analysis – Example 2

Problem:

Calculate the regression coefficient and obtain the lines of regression for x on y the following data.

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

Solution: This problem involves only two variable, x and y . This would require a simple linear regression equation to be fitted.

$$\text{Slope of } x \text{ on } y = b_{xy} = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\frac{\sum y^2}{n} - \bar{y}^2} =$$

$$\frac{47.7 - (4.0 * 11.0)}{125.0 - (11.0 * 11.0)} = 0.925$$

The intercept is calculated from

$$\bar{x} = m\bar{y} + C2 \Rightarrow 4 = 0.925(11) + C2 \Rightarrow C2 = -6.175$$

The Regression equation is:

$$x = 0.925y - 6.175$$

n	x	y	x ²	y ²	xy
1	1	9	1	81	9
2	2	8	4	64	16
3	3	10	9	100	30
4	4	12	16	144	48
5	5	11	25	121	55
6	6	13	36	169	78
7	7	14	49	196	98
Total	28	77	140	875	334
Mean	4.0	11.0	20.0	125.0	47.7

Simple Regression Analysis – Example 2

Correlation coefficient (R) is calculated using the formula

$$R_{xy} = \frac{\frac{\sum xy}{n} - \bar{x} \bar{y}}{\sqrt{\frac{\sum x^2}{n} - \bar{x}^2} * \sqrt{\frac{\sum y^2}{n} - \bar{y}^2}}$$

$$R_{xy} = \frac{47.7 - (4.0 * 11.0)}{\sqrt{20.0 - 4^2} * \sqrt{125 - 11^2}} = 0.925$$

Note; $R_{xy} = R_{yx}$ as both have the same expression

The coefficient of determination $R_{xy}^2 = 0.856$

This implies that 85.6% of the variation in x with changes in y and vice-versa can be explained by the equation $x = 0.925y - 6.175$

We can also state that linear equations are reasonably the best fit as $R_{xy} = 0.925$

Simple Regression Analysis – Example 3

Problem: The table below gives the results obtained from an experiment for the film thickness in millimetres of bread crust as a function of time in minutes that the bread specimen was at temperature. Fit a suitable regression line, also find the correlation coefficient and comment on the fit.

Time in minutes	20	30	40	60	70	90	100	120	150	180
Thickness (mm)	3.5	7.4	7.1	15.6	11.1	14.9	23.9	27.1	22.1	23.9

Solution: This problem involves only two variable, time and thickness which can be designated as x and y respectively. This would require a simple linear regression equation to be fitted. The data can be tabulated to calculate the Σy , Σx , Σxy , Σx^2 and Σy^2 values as shown in the next slide.

Simple Regression Analysis – Example 3

x = Time and y = Thickness					
n_i	x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1	20.0	3.5	400.0	12.3	70.0
2	30.0	7.4	900.0	54.8	222.0
3	40.0	7.1	1,600.0	50.4	284.0
4	60.0	15.6	3,600.0	243.4	936.0
5	70.0	11.1	4,900.0	123.2	777.0
6	90.0	14.9	8,100.0	222.0	1,341.0
7	100.0	23.9	10,000.0	571.2	2,390.0
8	120.0	27.1	14,400.0	734.4	3,252.0
9	150.0	22.1	22,500.0	488.4	3,315.0
10	180.0	23.9	32,400.0	571.2	4,302.0
Sum	860.0	156.6	98,800.0	3,071.2	16,889.0
Mean	86.0	15.7	9,880.0	307.1	1,688.9

$$\Sigma y = 156.6$$

$$\Sigma x = 860.0$$

$$\Sigma xy = 16,889.0$$

$$\Sigma x^2 = 98,800.0$$

$$\Sigma y^2 = 3,071.2$$

$$\bar{x} = 86.0$$

$$\bar{y} = 15.7$$

$$\Sigma x^2/n = 9880.0$$

$$\Sigma y^2/n = 307.1$$

$$\Sigma xy/n = 1688.9$$

$$\text{Slope of } y \text{ on } x = b_{yx} = \frac{\frac{\Sigma xy}{n} - \bar{x}\bar{y}}{\frac{\Sigma x^2}{n} - \bar{x}^2} = \frac{1688.9 - (86.0 * 15.7)}{9880.0 - (86.0 * 86.0)} = 0.1364$$

The intercept is calculated from $\bar{y} = m\bar{x} + C$
where m is the slope and C is the intercept

Substituting $15.7 = 0.1364(86.0) + C$; $C = 3.9696$ is the intercept

The Regression Equation is : $y = 0.1364x + 3.9696$

Simple Regression Analysis – Example 3

Correlation coefficient (R) is calculated using the formula

$$R_{xy} = \frac{\frac{\sum xy}{n} - \bar{x} \bar{y}}{\sqrt{\frac{\sum x^2}{n} - \bar{x}^2} * \sqrt{\frac{\sum y^2}{n} - \bar{y}^2}}$$
$$R_{xy} = \frac{1688.9 - 86.0 * 15.7}{\sqrt{9880.0 - 86.0^2} * \sqrt{307.1 - 15.7^2}} = 0.873$$

The coefficient of determination $R_{xy}^2 = 0.762$

This implies that 76.2% of the variation in film thickness (y) with changes in the time (x) can be explained by the equation $y = 0.1364x + 3.9696$.

We can also state that linear equation is reasonably the best fit for the experimental data to express relationship between film thickness and time since $R = 0.873$

Simple Regression Analysis – Example 4

Problem: Find the means of x and y variables and the coefficient of correlation between them from the following two regression equations: $2y - x - 50 = 0$ and $3y - 2x - 10 = 0$.

Solution: Given two regression equations.

$$2y - x - 50 = 0 \dots (1)$$

$$3y - 2x - 10 = 0 \dots (2)$$

Solving them simultaneously will provide the $\bar{x}$ and $\bar{y}$ (Mean of x and y).

$$\Rightarrow 2*(1) - (2) \Rightarrow y - 90 = 0, \text{ So } y = 90$$

$$\Rightarrow \text{Subs. in (1) gives: } 180 - x - 50 = 0, x = 130$$

So, $\bar{x} = 130$ and $\bar{y} = 90$ Answer for first part of the question

Simple Regression Analysis – Example 4

Out of the two regression equations, we can assume one of them to be for y on x and the other automatically becomes x on y

Let $2y - x - 50 = 0$ (1) be y on x, so this can be rearranged to

$$2y = 50 + x \text{ and } y = 0.5x + 25$$

$$\text{Hence } b_{y_x} = 0.5$$

Similarly $3y - 2x - 10 = 0$ (2) can be rearranged to

$$2x = 3y - 10 \text{ and } x = 1.5y - 5$$

$$\text{Hence } b_{x_y} = 1.5$$

$$\text{As } b_{y_x} = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\frac{\sum x^2}{n} - \bar{x}^2}; b_{x_y} = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\frac{\sum y^2}{n} - \bar{y}^2} \text{ and } R_{xy} = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\sqrt{\frac{\sum x^2}{n} - \bar{x}^2} * \sqrt{\frac{\sum y^2}{n} - \bar{y}^2}}$$

$$R_{xy} = \sqrt{b_{x_y} * b_{y_x}} = \sqrt{1.5 * 0.5} = 0.866$$

Simple Regression Analysis – Assignment

1. For 5 pairs of observations the following results are obtained, $\Sigma y = 25$, $\Sigma x = 15$, $\Sigma xy = 83$, $\Sigma x^2 = 55$ and $\Sigma y^2 = 135$. Find the equation of the regression lines and estimate the value of x on the 1st line when $y = 12$ and value of y on the 2nd line if $x = 8$.
2. The two regression lines are $3x + 2y = 26$ and $6x + 3y = 31$. Find the correlation coefficient.
3. In a laboratory experiment on correlation research study the equation of the two regression lines were found to be $2x - y + 1 = 0$ and $3x - 2y + 7 = 0$. Find the means of x and y . Also work out the values of the regression coefficient and correlation between the two variables x and y .
4. For the given lines of regression $3x - 2y = 5$ and $x - 4y = 7$. Find (i) Regression coefficients and (ii) Coefficient of correlation
5. Find the regression coefficients for the following data:

Age (Years)	43	21	25	42	57	59
Glucose Level (mg/dL)	99	65	79	75	87	81

Multiple Regression Models

In the case of the simple regression model, we assume only one independent variable. However, in real life examples, more than one independent variables exist. For example:

Agricultural yield = $f(\text{rainfall, soil fertility, technology adoption, pest damage})$

$y = f(x_1, x_2, x_3, x_4)$ with four independent variables
Where $y = \text{yield}$, $x_1 = \text{rainfall}$, $x_2 = \text{soil fertility}$,
 $x_3 = \text{technology adoption}$ and $x_4 = \text{pest damage}$

For e.g.: $y = 0.78 + 0.1 x_1 + 0.25 x_2 + 0.06 x_3 + 4.6 x_4$

So, when rainfall increases by 0.1 units, yield increases by one unit
and so on....

Multiple Regression Analysis

Multiple linear regression analysis (MLR) represents the logical extension of the simple case to two or more independent variables to estimate the significance of the dependent variable

Data Structure for Multiple Regression Analysis								
n	y	x ₁	x ₂	x ₃	x ₄	-	-	x _n
1	y _{1,1}	x _{1,1}	x _{2,1}	x _{3,1}	x _{4,1}	-	-	x _{n,1}
2	y _{1,2}	x _{1,2}	x _{2,2}	x _{3,2}	x _{4,2}	-	-	x _{n,2}
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
						-	-	-
Σn	Σy ₁	Σx ₁	Σx ₂	Σx ₃	Σx ₄	-	-	Σx _n

Reg. Eqn.: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 \dots + \beta_n x_n + e$
 where “e” is error term, β_0 is a constant representing the effect of variables not included in the model and $\beta_1, \beta_2, \beta_3 \dots \beta_n$ are regression coefficients associated with each independent variable

Multiple Regression Analysis - Examples

Evaluate the following dataset to fit a multiple linear regression model

y	140	155	159	179	192	200	212	215
x ₁	60	62	67	70	71	72	75	78
x ₂	22	25	24	20	15	14	14	11

The data can be tabulated as shown below with x_1^2 , x_2^2 , x_1y , x_2y and x_1x_2

n	y	x ₁	x ₂	x ₁ ²	x ₂ ²	x ₁ y	x ₂ y	x ₁ x ₂
1	140	60	22	3,600	484	8,400	3,080	1,320
2	155	62	25	3,844	625	9,610	3,875	1,550
3	159	67	24	4,489	576	10,653	3,816	1,608
4	179	70	20	4,900	400	12,530	3,580	1,400
5	192	71	15	5,041	225	13,632	2,880	1,065
6	200	72	14	5,184	196	14,400	2,800	1,008
7	212	75	14	5,625	196	15,900	2,968	1,050
8	215	78	11	6,084	121	16,770	2,365	858
SUM	1,452	555	145	38,767	2,823	101,895	25,364	9,859
MEAN	181.5	69.4	18.1	4,845.9	352.9	12,736.9	3,170.5	1,232.4

$$\text{With } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$$

Multiple Regression Analysis - Examples

To determine β_0 , β_1 , and β_2 :

Calculate the Regression Sums of squares, using the formulae:

- Reg. Sum of $x_1^2 = \Sigma x_1^2 = \Sigma x_1^2 - \{(\Sigma x_1)^2/n\} = 38,767 - \{(555)^2/8\} = 263.9$
- Reg. Sum of $x_2 = \Sigma x_2^2 = \Sigma x_2^2 - \{(\Sigma x_2)^2/n\} = 2,823 - \{(145)^2/8\} = 194.9$
- Reg. Sum of $x_1y = \Sigma x_1y = \Sigma x_1y - \{(\Sigma x_1 * \Sigma y)/n\} = 101,895 - \{(555 * 1,452)/8\} = 1,162.5$
- Reg. Sum of $x_2y = \Sigma x_2y = \Sigma x_2y - \{(\Sigma x_2 * \Sigma y)/n\} = 25,364 - \{(145 * 1,452)/8\} = -953.5$
- Reg. Sum of $x_1x_2 = \Sigma x_1x_2 = \Sigma x_1x_2 - \{(\Sigma x_1 * \Sigma x_2)/n\} = 9,859 - \{(555 * 145)/8\} = -200.4$

This is tabulated as shown below

n	y	x_1	x_2	x_1^2	x_2^2	x_1y	x_2y	x_1x_2
REG. SUM				263.9	194.9	1,162.5	-953.5	-200.4

Multiple Regression Analysis - Examples

Using this regression of sum of squares data, the regression constants can be calculated using the formulae:

$$\beta_1 = \frac{[(\Sigma x_2^2)(\Sigma x_1 y) - (\Sigma x_1 x_2)(\Sigma x_2 y)]}{[(\Sigma x_1^2)(\Sigma x_2^2) - (\Sigma x_1 x_2)^2]} = \frac{[(194.9)(1162.5) - (-200.4)(-953.5)]}{[(263.9)(194.9) - (-200.4^2)]} = \frac{35,489.85}{11,273.95} = 3.148$$

$$\beta_2 = \frac{[(\Sigma x_1^2)(\Sigma x_2 y) - (\Sigma x_1 x_2)(\Sigma x_1 y)]}{[(\Sigma x_1^2)(\Sigma x_2^2) - (\Sigma x_1 x_2)^2]} = \frac{[(263.9)(-953.5) - (-200.4)(1162.5)]}{[(263.9)(194.9) - (-200.4^2)]} = \frac{-18,663.65}{11,273.95} = -1.655$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}_1 - \beta_2 \bar{x}_2 = 181.5 - 3.148(69.4) - (-1.655)(18.2) = -6.850$$

Therefore the regression equation is given by:

$$y = -6.85 + 3.148x_1 - 1.655x_2$$

- $\beta_0 = -6.85 \Rightarrow$ When both independent variables, x_1 and x_2 are equal to zero, the mean value for y is -6.85.
- $\beta_1 = 3.148 \Rightarrow$ A one unit increase in x_1 is associated with a 3.148 unit **increase** in y , on average, assuming x_2 is held constant.
- $\beta_2 = -1.656 \Rightarrow$ A one unit increase in x_2 is associated with a 1.656 unit **decrease** in y , on average, assuming x_1 is held constant.

Sampling Statistic and Sizes

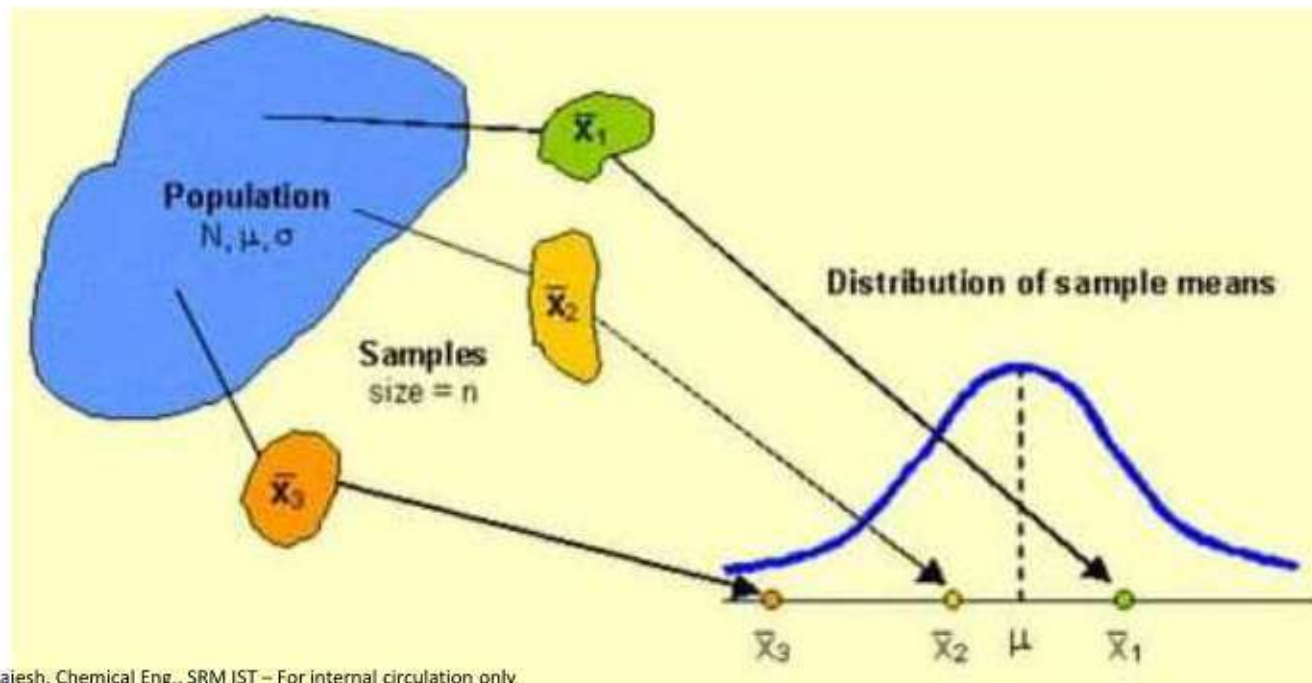
- A “**sample**” is a carefully chosen representative part of the “**population**” used to draw “**inference**” about the population.
- “**Parameter**” is a number that describe the property of entire population, whereas “**statistic**” is a number that describe the properties of a sample.

	Population	Sample
Size	N	n
Mean	μ	$\bar{x}$
Standard Deviation	σ	s
Variance	σ^2	s^2
Correlation Coefficient	ρ	R
Observations	$X_1, X_2, \dots, X_N$	$x_1, x_2, \dots, x_n$
Properties	Parameter $\Theta (N, \mu, \sigma, \sigma^2, \rho)$	Statistic $\Theta^* (n, \bar{x}, s, s^2, r)$

- Sample sizes can be either small ($n \leq 30$) or large ($n > 30$)

Sampling Sizes and Distributions

- A **sampling distribution** is a probability distribution of a statistic that is obtained through repeated sampling of a population.
- It describes a range of possible outcomes for a statistic, such as the mean or mode obtained through repeated sampling of a population.
- Illustration below shows various means that may be obtained from a population may be distributed and their probability.

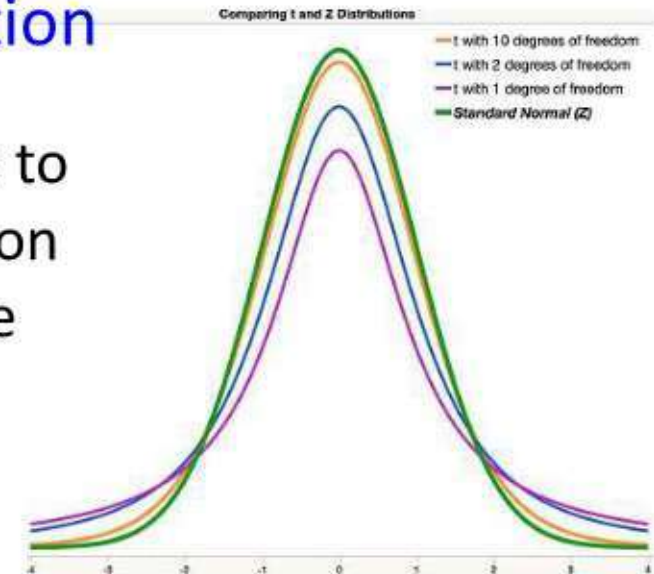


Sampling Distributions

- The **sampling distributions** for small sample sizes are:
 - Student's t-distribution ($n \leq 30$)
 - Chi-Square (χ^2) Distribution (Both $n \leq 30$ and $n > 30$)
 - F-Distribution ($n \leq 30$)

Student's T-Distribution

- The t-distribution describes the standardized distances of sample means to the population mean when the population standard deviation is not known, and the observations come from a normally distributed population. Also $n \leq 30$.
- Please note that in contrast, the standard normal or z-distribution assumes that the population standard deviation is known with $n > 30$.



1. Student's T-Distribution

- Definition: If $x_1, x_2, \dots, x_n$ is a sample of size “n” from a normal population with unknown μ and unknown variance σ^2 and if

$$\bar{x} = \frac{\sum x_i}{n} \text{ and } s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}, \text{ then the statistic } t, \text{ is given by}$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1} \text{ df}$$

Where $\sim$ means follows and df means degrees of freedom

- The t-test is used to compare the:
 - A. One-sample t-test for Sample Mean and Population Mean
 - B. Paired t-test for Sample Means before and after any change inducing event, say a treatment of the population
 - C. Two-Sample t-test – for means of two different samples
 - D. Difference between Sample Correlation Coefficient and Population Correlation Coefficient

A. One-Sample T-Test Example

- In this type of t-test, the null hypothesis is based on the assumption that the population mean is equal to a specified value μ based on a sample mean $\bar{x}$.
- For e.g. $H_0: x = \mu$ and $H_A: x \neq \mu$
- In other words, the population mean and sample mean are compared
- For a one-sample t-test, the following formula for the t-score is used.

$$t\text{-test} = \frac{(\bar{x} - \mu)}{s / \sqrt{n}}$$

- Another t-score called as the **t-critical** value can be obtained from tables or from any statistical software. MS Excel can also be used for this. The **t-test** and **t-critical** value are compared to test the hypothesis as explained in the example that follows

One-Sample T-Test - Example

A retail store wants to improve sales. Historical sales data indicate that the average sale was Rs. 100 per transaction. After training the sales force, recent sales data (taken from a sample of 25 salespeople) indicates an average sale of Rs. 130, with a standard deviation of Rs. 15. Did the training work? Assume that the level of significance is 5%. Also assume that the t-critical value is 1.71.

Solution

Step 1. Framing Null and Alternate Hypothesis.

$H_0: x = \mu$, As Average sales, $x = 100$, $\mu = 100$

$H_A: x \neq \mu$, As Average sales, $x = 100$, $\mu > 100$

Step 2. Choose test statistic- In this problem, the sample size, $n = 25$ with $n < 30$ and small sample size. Also, the population standard deviation is not known. Hence we will be using a t statistic.

One-Sample T-Test - Example

Step 3. Level of significance is given, so $\alpha = 0.05$

Step 4. Calculate t-test,

$$\text{t-test} \Rightarrow t - \text{test} = \frac{(\bar{x} - \mu)}{s/\sqrt{n}} = \frac{(130 - 100)}{15/\sqrt{25}} = 10$$

Step 5. Calculate t-critical, but it is given here as t-critical = 1.7

Step 6. Decision Making about the null hypothesis

Reject a Null Hypothesis, if t-test > t-critical and accept if t-test < t-critical.

Here t-test > t-critical. So, Null Hypothesis is rejected and alternate is accepted, i.e. $H_A: x \neq \mu, \mu > 100$. So there is an improvement in the sales and the training worked.

B. Paired T-Test

- A paired t-test can be used to compare two samples when the:
 - Samples are dependent or related in some way with each other OR
 - Samples are repeated
- An example is:
 - Measurement of test scores of the same set of students before and after a special coaching class
 - So, the mean values of the scores **before and after** the class are compared to check whether the coaching is effective.
- In this type of t-test, the following formula is used for calculating a t-score:

$$\text{Observed } t = t_o = \frac{\bar{d}}{\sqrt{s_d^2/n}} \text{ where } \bar{d} = \frac{\sum d_i}{n} \text{ and } s_d^2 = \frac{[\sum d_i^2 - \{(\sum d_i)^2/n\}]}{n-1}$$

- Critical t, t_c is either provided or calculated from tables using the significance value, α
- Also, if $t_o < t_c$, the null hypothesis is valid and *vice-versa*

Paired T-Test - Example

Consider 5 people working in an office such that they undergo a training as shown below. Let the expected t , i.e. $t_e = 2.78$ be the outcome after training. Check whether the training is effective.

Training Test Marks				
	Before	After	Difference (d)	d^2
1	10	15	5	25
2	15	15	0	0
3	17	12	-5	25
4	12	17	5	25
5	14	11	-3	9
SUM			$\Sigma d = 2$	$\Sigma d^2 = 84$

Formulating the Hypothesis

- H_0 : Training is not effective (Negative outcome is H_0)
- H_A : Training is effective

- Here, the before and after effect of the training is compared. This requires a paired t-test. In this test:

- $t = \frac{\bar{d}}{\sqrt{s_d^2/n}}$ where $\bar{d} = \frac{\Sigma d_i}{n} = \frac{2}{5} = 0.4$ & $s_d^2 = \frac{[\Sigma d_i^2 - \{(\Sigma d_i)^2/n\}]}{n-1} = \frac{[84 - \{\frac{4}{5}\}]}{4} = 20.8$

- Subs: Observed $t_0 = \frac{0.4}{\sqrt{20.8/5}} = 0.2$. Also $t_0 = 0.2 > t_e = 2.78$

- Hence the Null Hypothesis is valid and training is not effective!

One-Sample and Paired T-Test - Assignment

Problem:

1. Check whether the mean sample value differs significantly from that of the population for tabulated student scores. Let the critical t-value be 2.201.

Student	1	2	3	4	5	6	7	8	9	10	11	12
Scores	28	29	35	37	32	26	37	39	22	29	36	38

2. An instructor administers two exams to a class of students. The scores are tabulated below. Check if the exams are equally difficult.

Student	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Exam 1 Score	63	65	56	100	88	83	77	92	90	84	68	74	87	64	71	88
Exam 2 Score	69	65	62	91	78	87	79	88	85	92	69	81	84	75	84	82

C. Two-Sample Means T-Test

- It tests the null hypothesis whether two sample means $\bar{x}_1$ and $\bar{x}_2$ are equal
- Consider two samples 1 and 2 with distinct means, $\bar{x}_1$ and $\bar{x}_2$

	Size	Mean	Observations
Sample 1	n_1	$\bar{x}_1$	$x_1, x_2 \dots x_n$
Sample 2	n_2	$\bar{x}_2$	$x_1, x_2 \dots x_n$

- The hypothesis is that $H_0: \bar{x}_1 = \bar{x}_2$ and $H_A: \bar{x}_1 \neq \bar{x}_2$
- The following formula is used for calculating t-score in this type of t-test
- With $\bar{x}_1 = \frac{\sum x_1}{n}$ and $\bar{x}_2 = \frac{\sum x_2}{n}$
- and $s^2 = \frac{\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}$
- $$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1 + n_2 - 2} \text{ df}$$
- The t_o and t_e can be compared to infer about the hypothesis

D. Sample and Population Correlation Coefficients T-Test

- To test for the difference between sample correlation coefficient (r) and population correlation coefficient (ρ), the following t-test with t-score formula is used:
- $$t = \frac{r}{\sqrt{1-r^2}} \sqrt{n-2} \sim t_{n-2} df$$
- Where t_{n-2} is t with $n-2$ degrees of freedom
- Comparing t_o and t_e , appropriate inference maybe made for the null hypothesis

2. Chi-Square (χ^2) Distribution

The square of a standard normal variate is known as chi-square with one degree of freedom

Symbolically $x \approx N(\mu, \sigma^2)$, then $Z = \frac{x-\mu}{\sigma} \approx N(0,1)$

$Z^2 = \left(\frac{x-\mu}{\sigma}\right)^2$ is chi-square with degree of freedom, $df = 1$

$$\chi^2 = \sum \left(\frac{x - \mu}{\sigma}\right)^2 \approx \chi_n^2$$

Applications of Chi-square distributions

1. To test the goodness of fit, *i.e.* validation of the model
2. To test the independence of attributes, *i.e.* association between variables

These applications are discussed in the next slides

χ^2 Distribution – Example 1

Goodness of Fit (Validation of Model)

Problem: A official Tamil Nadu population census was conducted in 2021 and it was reported that in Potheri town, 60% of families have only one child, 28% have two children, and 12% have three or more children. To verify this, a student of SRM had decided to conduct his own survey in Potheri town and has found that out of the 1290 families in the town, 730 had one children and 380 had two children. Determine whether his data supports the results of the census. A significance level, $\alpha = 0.05$ may be assumed.

Solution:

Stating the null and alternative hypotheses:

H_0 : In student's survey, the data does not match with the census

H_A : In the student's survey, the data does match with the census

χ^2 Distribution – Example 1

Goodness of Fit (Validation of Model)

Solution: A χ^2 goodness of fit test is appropriate because we are examining the distribution of a single variable.

The population data may be tabulated to calculate the observed chi-square value

Number of Children	Observed (O_i)	Expected (E_i) as per census	$O_i - E_i$	$(O_i - E_i)^2$	$(O_i - E_i)^2/E_i$
One	730	$0.60 \times 1290 = 774$	-44	1936	2.5
Two	380	$0.28 \times 1290 = 361$	19	361	1
$\geq$ Three	180	$0.12 \times 1290 = 155$	25	625	4.03
SUM	1290	1290	0	2922	7.53
	ΣO_i	ΣE_i	$\Sigma(O_i - E_i)$	$\Sigma(O_i - E_i)^2$	$\Sigma(O_i - E_i)^2/E_i$

The observed chi-square value is: $\chi_o^2 = \Sigma(O_i - E_i)^2/E_i = 7.53$.

χ^2 Distribution – Example 1

Goodness of Fit (Validation of Model)

To calculate the expected chi-square value, the degrees of freedom is given by: $df = 3 - 1 = 2$

With $df = 2$ and $\alpha = 0.05$, The expected chi-square value observed from the table is: $\chi_e^2 = 5.99$

Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09

As $\chi_o^2 > \chi_e^2$, the H_A is valid and fit is a good fit

So, the student's survey data is in concurrence with the census

χ^2 Distribution – Example 2

Testing Association Between Attributes

- The Chi-Square Test for Association is used to determine if there is any association between two or more categorical values.
- It is used to compare the observed and expected results and identify whether the disparity that exists between actual and predicted data is due to chance or to a link between the variables under consideration.
- As a result, the chi-square test is an ideal choice for aiding in our understanding and interpretation of the connection between our two categorical variables.
- It is a hypothesis test of independence.
- The null hypothesis is that the variables are not associated, *i.e.*, independent. The alternate hypothesis is that the variables are associated.

χ^2 Distribution – Example 2

Testing Association Between Attributes

Problem: Test for association between the two attributes as tabulated for a survey among smokers and non-smokers.

Observed Distribution Frequencies

OBSERVED	Lung Ailment	No Lung Ailment	Total
Smokers	450	70	520
Non-smokers	60	420	480
Total	510	490	1000

Note: A cursory glance at the data is enough to suggest that there is strong correlation between smoking and lung ailment. But, let us try to prove this statistically with the chi-square test.

χ^2 Distribution – Example 2

Testing Association Between Attributes

Step 1: Framing Hypothesis

H_o = There is no association between smoking and lung cancer

H_A = There is an association between smoking and lung cancer

Step 2: Constructing a table of expected frequencies

- A table must be constructed that identifies **expected** distribution frequencies (for comparison against observed)
- Expected frequencies are calculated with the formula:

Expected frequency = (Row total $\times$ Column total) $\div$ Grand total

OBSERVED	Lung Ailment	No Lung Ailment	Total
Smokers	450	70	520
Non-smokers	60	420	480
Total	510	490	1000



EXPECTED	Lung Ailment	No Lung Ailment
Smokers	$520 \times 510 / 1000 = 265.2$	$520 \times 490 / 1000 = 254.8$
Non-smokers	$480 \times 510 / 1000 = 244.8$	$480 \times 490 / 1000 = 235.2$

χ^2 Distribution – Example 2

Testing Association Between Attributes

Step 3: Apply the chi-squared formula

The formula used to calculate a statistical value for the chi-squared test is: $\chi^2 = \sum \frac{(O-E)^2}{E}$, where: O = Observed and E = Expected frequency

CHI-SQUARE FORMULA		Lung Ailment	No Lung Ailment
Smokers	O	450.0	70.0
	E	265.2	254.8
	$(O-E)^2/E$	128.8	134.0
Non-smokers	O	60.0	420.0
	E	244.8	235.2
	$(O-E)^2/E$	139.5	145.2

$$\chi_o^2 = \sum \frac{(O-E)^2}{E} = 128.8 + 134.0 + 139.5 + 145.2 = 547.5$$

χ^2 Distribution – Example 2

Testing Association Between Attributes

Step No. 4: Identify the Degrees of Freedom (df)

In order to determine if the chi-squared value is statistically significant a degree of freedom must first be identified. The degree of freedom is a mathematical restriction that designates what range of values fall within each significance level. The degree of freedom is calculated from the table of frequencies according to the following formula:

$$df = (R - 1) (C - 1)$$

Where: R = number of rows ; C = number of columns

Here the degrees of freedom is $df = (R-1)(C-1) = (2-1)(2-1) = 1$

χ^2 Distribution – Example 2

Testing Association Between Attributes

Step No. 5: Identify the p Value

- The final step is to apply the value generated to a chi-squared distribution table to determine if results are statistically significant. A value is considered significant if there is less than a 5% probability ($p < 0.05$) the results are attributable to chance

Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09

- For $p = 0.05$ and $df = 1$, the critical value, *i.e.* expected chi-square distribution value is 3.841.
- So, when $df = 1$, a value of greater than 3.841 is required for results to be considered statistically significant ($p < 0.05$).

χ^2 Distribution – Example 2

Testing Association Between Attributes

Step No. 6: Final Results

Here $\chi_o^2 = 547.5 > \chi_e^2$ (critical value) = 3.841

Hence, the difference between observed and expected frequencies are statistically significant

As the results are statistically significant, the null hypothesis is rejected and the alternate hypothesis accepted.

H_o = There is no association between smoking and lung ailment

H_A = There is an association between smoking and lung ailment

There is a strong association between smoking and lung ailment.

χ^2 Distribution – Assignment

Goodness of Fit Test

1. A candy manufacturing company decides to test the choice of flavour of candy amongst 1000 students. The raw data for type of candy flavour preferred by children is tabulated. The company had also hypothesized (expects) that there will be equal preferences for the flavours. Use the Chi-square goodness of fit test to check whether the hypothesis hold true. Let the Chi-square value with $\alpha = 0.05$ and 4 degrees of freedom be 9.488.

Candy	Numbers	
Flavour	Observed	Expected
Apple	180	200
Lime	250	200
Cherry	120	200
Pineapple	225	200
Grape	225	200
Sum	1000	1000

χ^2 Distribution – Assignment

Testing for Association Between Attributes

1. Tabulated is the raw data of high school boys and girls and their respective behavioural issues (or lack thereof) in a class. Use a chi-square test to determine whether there is a relationship between gender and being troublesome at school. Let the significance level be set at 0.05 $\chi_e^2 = 3.84$. (Clue: $\chi_o^2 = 1.87$).

	Troublesome	
	Yes	No
Boys	46	71
Girls	37	83

2. The survey of the voter preference for a party and their gender are tabulated. To check whether the gender has anything to do with political party preference, conduct a chi-square test. Let significance level be 0.05. (Clue: $\chi_o^2 = 9.837$)

	Voting Results		
	Party A	Party B	Party C
Male	100	70	30
Female	140	60	20

F-Distribution

- F test is a statistical test that is used in hypothesis testing to check whether the **variances of two populations or two samples** are equal or not.
- To use F distribution under the null hypothesis, it is important to determine the mean of the two given observations at first and then calculate the variance.
- In an f test, the data follows an f distribution. This test uses the f-statistic to compare two variances by dividing them.
- F-statistic = σ_1^2 / σ_2^2 for large and = s_1^2 / s_2^2 for small samples
- Null Hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$
- Alternate Hypothesis: $H_A: \sigma_1^2 \neq \sigma_2^2$
- Decision Criteria: If the f-statistic $\neq$ f-critical value then H_0 is rejected

F-Distribution – Example 1

- **Problem:** A research team wants to study the effects of a new drug on insomnia. 8 tests were conducted with a variance of 600 initially. After 7 months 6 tests were conducted with a variance of 400. At a significance level of 0.05 was there any improvement in the results after 7 months? Given F-critical value = 4.88
- **Solution:** As variance needs to be compared, the f-test is used.
- $H_0: s_1^2 = s_2^2$ and $H_A: s_1^2 \neq s_2^2$
- $n_1 = 8, df_1 = 8 - 1 = 7, s_1^2 = 600$
- $n_2 = 6, df_2 = 6 - 1 = 5, s_2^2 = 400$
- F-test formula: $= s_1^2 / s_2^2$ (for small samples) $= 600/400 = 1.5$
- Given from F-table the critical value $F(0.05, 7, 5) = 4.88$
- As $1.5 < 4.88$, the null hypothesis cannot be rejected and there is not enough evidence to conclude that there was an improvement in insomnia after using the new drug.

F-Distribution – Example 2

- Problem:** For the two sets of sample data check whether they have been drawn from the normal population with the same variance, s^2 .

	1	2	3	4	5
Sample 1	14	12	11	14	9
Sample 2	15	17	28	10	

- Solution:** Tabulate as shown below with $n_1 = 5$ and $n_2 = 4$

	x	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	14	2	4
2	12	0	0
3	11	-1	1
4	14	2	4
5	9	-3	9
Sum	60	0	18
Mean	12	0	3.6

	y	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
1	15	-2.5	6.25
2	17	-0.5	0.25
3	28	10.5	110.25
4	10	-7.5	56.25
Sum	70	0	173
Mean	17.5	0	43.25

F-Distribution – Example 2

- Null Hypothesis: $H_o = s_x^2 = s_y^2$ and no significant difference
- Alternate Hypothesis: $H_A = s_x^2 \neq s_y^2$ and there is a significant difference
- So, $s_x^2 = (x_i - \bar{x})^2 / (n-1) = 18/4 = 4.5$ and
- $s_y^2 = (y_i - \bar{y})^2 / (n-1) = 173/4 = 57.67$
- Observed $F = F_o = s_y^2 / s_x^2 = 57.67 / 4.5 = 12.81$
- Expected $F = F_e = 6.59$ for $n_1-1 = 4$ and $n_2-1 = 3$ and $\alpha = 0.05$ (from tables)
- As $F_o > F_e$, there is a significant difference between the means of the two samples

F-Distribution - Assignment

1. The pizza delivery times of two cities are given below.

City 1: Number of delivery times observed = 28, Variance = 38

City 2: Number of delivery times observed = 25, Variance = 83

Check if the delivery times of city 1 are lesser than city 2 at a 0.05 alpha level. Let f-critical value be 0.5181. (Answer: H_0 is rejected)

2. A toy manufacturer wants to get batteries for toys. A team collected 41 samples from supplier A and the variance was 110 hours. The team also collected 21 samples from supplier B with a variance of 65 hours. At a 0.05 alpha level determine if there is a difference in the variances. Let f-critical be 2.287. (Answer: H_0 is not rejected)

Classification of Data

One Way Classification

One dependent and one independent variables or factors

E.g.: Marathon Race Time Vs. Shoe brand (Nike, Adidas *etc.*)

Two Way Classification

One dependent and two independent variables or factors

E.g.: Marathon Race Time Vs. Shoe brand (Nike, Adidas *etc.*) and Age group of the runners (Junior, Senior *etc.*)

Three Way Classification

One dependent and three independent variables or factors

E.g.: Marathon Race Time Vs. Shoe brand (Nike, Adidas *etc.*), Age group of the runners (Junior, Senior *etc.*) and Weight of the runners

One-Way Classification of Data

One-way classification has an experimental result layout in which sample observations are classified (grouped) by only a single criterion. It provides the simplest data structure.

Consider a scenario where 3 different drugs are administered to different subjects and the relief time (in minutes) of the symptoms are noted as tabulated. This is an example of one-way classification of data with one dependent and one independent variable.

Relief Time = $f(\text{Drug Type})$

So Subject 1 taking Drug D1 gets relief in 15 minutes

Relief Time = $f(\text{Drug})$			
	Drug		
	D1	D2	D3
1	15	15	12
2	20	25	17
3	25	30	13
4	18	32	20
5	20	35	25
6	22	40	28

Two-Way Classification of Data

Two-way classification has two independent variables.

Consider a scenario where 3 different drugs are administered to different subjects and the relief time (in minutes) of the symptoms along with the age of the subjects are tabulated. This is an example of two-way classification of data with one dependent and two independent variable.

Relief Time = $f(\text{Drug Type, Age of Subject})$

Relief Time = $f(\text{Drugs, Age})$				
	Age	Drug		
	(Years)	D1	D2	D3
1	< 14	25	22	20
2	14 - 35	18	17	15
3	35 - 60	20	22	21
4	60 - 80	35	38	42
5	> 80	62	70	65

Three-Way Classification of Data

Three-way classification has three independent variables.

Consider a scenario where 3 different drugs (D1, D2 and D3) are administered to different subjects and the relief time (in minutes) of the symptoms along with the age of the subjects and time of administration of the drug are tabulated. This is an example of three-way classification of data with one dependent and three independent variable.

Relief Time = $f(\text{Age, Drug, Time of administration})$

Here the codes, S, T, U and V denote time of administration of the drug

Relief Time = $f(\text{Drugs, Age, Time of Adm})$				
	Age	Drug		
	(Years)	D1	D2	D3
1	< 14	S (25)	T (22)	V (20)
2	14 - 35	T (18)	U (17)	S (15)
3	35 - 60	U (20)	V (22)	T (21)
4	> 60	V (35)	S (38)	U (42)
S = 4 AM, T = 8 AM, U = 12 Noon, V = 4 PM				
Relief Time is provided in the brackets				

Analysis of Variance (ANOVA)

- Analysis of Variance (ANOVA) is a statistical formula used to compare variations across the **means of different** groups
- Consider a diabetes drug clinical trial with different groups (control, experimental 1, experimental 2 etc.) ANOVA helps to compare the means blood sugar levels of the groups to determine if they are statistically different or similar thereby establishing the efficacy of the drug
- The systematic procedure for achieving this is termed Analysis of Variance (ANOVA)
- The outcome of ANOVA is the 'F statistic'
- ANOVA was developed by statistician and evolutionary biologist, Ronald Fisher
- T-test is used for testing for significant differences between two means, but ANOVA is used for three or more means

ANOVA – Example 1

- **Problem:** Four coders are tested for their ability to write codes based on 4 model assignments. The ability to code may be based on the duration taken to execute the assignment. Hence, the time taken by coders for the assignments are tabulated below. Test whether there is any significant differences on the ability of the coders.

	Time Taken to Code (Hours)		
Coder	Model 1	Model 2	Model 3
1	8	7	6
2	5	4	2
3	6	5	3
4	4	5	5

- **Solution:**

Here R = Number of Rows = 4 and C = Number of Columns = 3

The total across rows and columns are calculated and tabulated as shown in the next slide

ANOVA – Example 1

	Time Taken to Code (Hours)				
Coder	Model 1	Model 2	Model 3	Total	
1	8	7	6	21	ΣR_1
2	5	4	2	11	ΣR_2
3	6	5	3	14	ΣR_3
4	4	5	5	14	ΣR_4
Total	23	21	16	60	
	ΣC_1	ΣC_2	ΣC_3	Grand Total	

Steps for the solution:

1. The Total Number of Observations= $N = R * C = 12$

2. The Grand Total = $G = \Sigma R_i = \Sigma C_i = 60$

3. Correction Factor = $CF = G^2/N = 60^2/12 = 300$

4. Total Sum of Squares =

$$\Sigma y_{ij}^2 - CF = \{(8^2+7^2+6^2)+(5^2+4^2+2^2)+(6^2+5^2+3^2)+(4^2+5^2+5^2)\} - 300 = 330 - 300 = 30$$

ANOVA – Example 1

5. Sum of Squares between Rows = Sum of Squares between

$$\text{Coders} = \sum \frac{R_i^2}{C} - CF = \frac{R_1^2 + R_2^2 + R_3^2 + R_4^2}{3} - CF = \frac{21^2 + 11^2 + 14^2 + 14^2}{3} - 300 = 318 - 300 = 18$$

6. Sum of Squares between Columns = Sum of Squares between

$$\text{Models} = \sum \frac{C_j^2}{R} - CF = \frac{C_1^2 + C_2^2 + C_3^2}{4} - CF = \frac{23^2 + 21^2 + 16^2}{4} - 300 = 306.5 - 300 = 6.5$$

7. Error Sum of Squares = Total Sum of Squares – Sum of Squares between Rows – Sum of Squares between Columns = 30 – 18 – 6.5 = 5.5

Note Error Sum of Squares is always supposed to be > 0. Here it is 5.5 > 0 and therefore found to be true.

ANOVA – Example 1

8. ANOVA Table Generation

	C_1	C_2	C_3	$C_4 = C_3/C_2$	$F_o = C_4/\text{Error}$	F_e (Provided in the Problem)
	Sources of Variation	Degrees of Freedom	Sum of Squares	Mean Sum of Squares	$F_{\text{observed}} (F_o)$	$F_{\text{expected}} (F_e)$
R-1	Between Coders (Rows)	$(R - 1) = (4 - 1) = 3$	18	$18/3 = 6$	$6/0.91 = 6.59$	4.76
C-1	Between Models (Columns)	$(C - 1) = (3 - 1) = 2$	6.5	$6.5/2 = 3.25$	$3.25/0.91 = 3.57$	5.14
(R-1)(C-1)	Error	$(N - 1) - (R - 1) - (C - 1) = 11 - 3 - 2 = 6$	5.5	$5.5/6 = 0.91$		
RC-1	Total	$(N - 1) = (12 - 1) = 11$	30			

The F_e value may be directly provided in the problem statement.

If not, it can also be deduced from the F-statistic table as described in the next slide.

ANOVA – Example 1

8. ANOVA Table Generation.. Continued

- The F_e can be obtained from the F-statistic table. If the $\alpha = 0.05$ (5% probability that the Null hypothesis is not true), then we can look up the corresponding F-static table for $\alpha = 0.05$
- For Between Rows: F_e with $df1 = R - 1 = 3$ and $df2 = R + C - 1 = 4 + 3 - 1 = 6$, With this, $F_e = 4.76$
- For Between Columns: F_e with $df1 = C - 1 = 3 - 1 = 2$ and $df2 = R + C - 1 = 6$, With this, $F_e = 5.14$

F-table of Critical Values of $\alpha = 0.05$ for $F(df1, df2)$																			
	DF1=1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
DF2=1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93

ANOVA – Example 1

9. Null Hypothesis Formulation (2 Hypothesis)

H_0 : There is no significant difference in the time taken to write the codes by the different coders

H_0 : There is no significant difference in the time taken to write the codes for the different models

10. Tabulating the Data

	Coders	Models
$F_{\text{observed}} = F_o$	6.59	3.57
$F_{\text{expected}} = F_e$	4.76	5.14

11. Inference: From the tabulated results data

Between Coders: $F_o (6.59) > F_e (4.76)$, we can infer that there is **a significant difference** between coders for the assignment

Between Models: $F_o (3.57) < F_e (5.74)$, we can infer that there is **no significant difference** in coding the models

ANOVA – Example 2

- **Problem:** Four workmen are allowed to work with 4 different machines in a factory. The units they are able to produce with each machine are tabulated below. Analyse the data to check whether there is any significant differences on the ability (i) of the machines and (ii) the workmen to produce the units. Let $F(3,9) = 3.86$ (F_{expected})

	No. of Units Produced			
Worker	Machine 1	Machine 2	Machine 3	Machine 4
A	14	17	22	11
B	15	19	24	18
C	17	13	20	14
D	12	10	9	8

ANOVA – Example 2

- **Solution:**

Here R = Number of Rows = 4 and C = Number of Columns = 4

The total across rows and columns are calculated as shown

	No. of Units Produced					
Worker	Machine 1	Machine 2	Machine 3	Machine 4	Total	
A	14	17	22	11	64	ΣR_1
B	15	19	24	18	76	ΣR_2
C	17	13	20	14	64	ΣR_3
D	12	10	9	8	39	ΣR_4
Total	58	59	75	51	243	
	ΣC_1	ΣC_2	ΣC_3	ΣC_4	Grand Total	

1. Total number of observations = $N = R * C = 4 * 4 = 16$
2. Grant Total = $G = \Sigma R_i = \Sigma C_i = 243$
3. Correction Factor = $CF = G^2/N = (243)^2/16 = 3690.56$

ANOVA – Example 2

4. Total Sum of Squares = $\sum y_{ij}^2 - CF = \{(14^2+17^2+22^2+11^2)+(15^2+19^2+24^2+18^2)+(17^2+13^2+20^2+14^2)+(12^2+10^2+9^2+8^2)\} - 3690.56 = 4019 - 3690.56 = 328.44$
5. Sum of Squares between Rows = Sum of Squares between Coders = $\sum \frac{R_i^2}{C} - CF = \frac{R_1^2+R_2^2+R_3^2+R_4^2}{4} - CF = \frac{64^2+76^2+64^2+39^2}{4} - 3690.56 = 181.69$
6. Sum of Squares between Columns = Sum of Squares between Models = $\sum \frac{C_j^2}{R} - CF = \frac{C_1^2+C_2^2+C_3^2+C_4^2}{4} - CF = \frac{58^2+59^2+75^2+51^2}{4} - 3690.56 = 77.19$
7. Error Sum of Squares = Total Sum of Squares – Sum of Squares between Rows – Sum of Squares between Columns = $328.44 - 181.69 - 77.19 = 69.56 > 0$

ANOVA – Example 2

8. ANOVA Table Generation

	C_1	C_2	C_3	$C_4 = C_3/C_2$	$F_o = C_4/\text{Error}$	F_e (Provided in the Problem)
	Sources of Variation	Degrees of Freedom	Sum of Squares	Mean Sum of Squares	$F_{\text{observed}} (F_o)$	$F_{\text{expected}} (F_e)$
R-1	Between Coders (Rows)	$(R - 1) = 3$	181.69	60.56	7.84	3.86
C-1	Between Models (Columns)	$(C - 1) = 3$	77.19	25.73	3.33	3.86
(R-1)(C-1)	Error	$(N - 1) - (R - 1) - (C - 1) = 9$	69.56	7.73		
RC-1	Total	$(N - 1) = 15$	328.44			

The F_e value is provided in the problem statement.

ANOVA – Example 2

9. Null Hypothesis Formulation (2 Hypotheses)

H_o : There is no significant difference in the production between the different workmen

H_o : There is no significant difference in the production time between the different machines

10. Tabulating the Data

	Workmen	Machines
$F_{\text{observed}} = F_o$	7.84	3.33
$F_{\text{expected}} = F_e$	3.86	3.86

11. Inference: From the tabulated results data

Between Workmen: F_o (7.84) > F_e (3.86), we can infer that there is **a significant difference** in the production time between workmen

Between Machines: F_o (3.33) < F_e (3.86), we can infer that there is **no significant difference** in the production time between machines